

Kaveen Sheron Jayasinghe

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About Me

An open-source enthusiast currently contributing to Oppia, with a passion for building secure software and deploying scalable systems that serve the greater good. I thrive on collaborating with others and believe the best way to learn is by doing.

Professional Experience

Training Co-Lead, Oppia (Google-backed Open-source Foundation) 2026 – Present

Angular, Python, Playwright

- Oppia is a learning tool that enables anyone to easily create and share interactive activities ("explorations") which try to recreate the experience of one-on-one tutoring, along with much more functionality.
- I contribute to the progress of this project by mentoring and onboarding new contributors, developing full-stack features, writing e2e frontend and backend tests, and coordinating quarterly project planning.

Associate Software Engineer, ictfromabc.com 2025 – Present

Laravel, SvelteKit, Flutter

- Contributed to developing and maintaining a large-scale LMS with an admin console, automating and digitalizing workflows for one of the largest tutoring institutes in Sri Lanka, serving 55,000+ users.
- Developed features including payment integration, user reviews with admin approval workflows, automated password recovery, and admin audit trails; improved platform reliability by writing backend tests and expanding CI workflows.

Open-source Contributor, WSO2 (Hacktoberfest) 2025

- [asgardeo-auth-react-sdk#305](#), [docs-is#5616](#), [identity-apps#9356](#), [identity-apps#10268](#).

Education

University of Moratuwa, Sri Lanka 2024 – Present

B.Sc. Engineering (Hons) in Computer Science & Engineering.
Specializing in Cyber Security.
CGPA 3.24 / 4.00

St. Joseph's College, Colombo-10 2009 – 2023

General Certificate Exam Advanced Level, Physical Sciences stream. Z-Score 2.2905, with 3 Distinction passes.
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General Certificate Exam Ordinary Level, 9 Distinction passes.

Projects

Prypiatos — Smart Energy Management System

[Prypiatos/repositories](#)

Terraform, Ansible, Kubernetes (k3s), Argo CD, AWS (EC2, EBS), MQTT, Kafka, WebSockets

- Prypiatos is a smart energy management system that applies machine learning to surface actionable insights and optimize energy usage.
- Provisioned a 3 node self managed k3s cluster on AWS with Terraform, defining EC2 instances and EBS storage as code, and automated cluster bootstrap with Ansible, covering control plane setup and worker node joining end to end.
- Implemented a GitOps deployment workflow with Argo CD, continuously syncing Git managed manifests with live cluster state
- Collected real-time energy data from ESP32 based edge nodes and streamed it through a Kafka pipeline for consumption by backend services, delivering live updates to the frontend via WebSockets.

Cross-Lingual Speech & Identity Synthesis System

[sheronjay/CrossLingual-FaceDub](#)

Python, WhisperX, NLLB-200, Coqui TTS, Wav2Lip, Voice Conversion

- Built an end-to-end English-to-Sinhala dubbing system that tries to preserve the original speaker's vocal identity and lip synchronization.
- Engineered a Python pipeline that transcribes audio with WhisperX, translates it via NLLB-200, romanizes the output, and synthesizes speech using a Coqui Sinhala TTS model, then applies Wav2Lip to align video with the dubbed audio.

Ohana Note — Secure Message-Sharing Platform with a Game Interface

[sheronjay/Ohana-Note](#)

KAPLAY, Spring Boot, MySQL, Docker Compose

- Built an end-to-end encrypted message-sharing platform where a user encrypts a message with a key shared privately with their partner, who then uses it to decrypt the message; wrapped in a playful game interface. Originally built for Valentine's Day as a fun side project, it has since been used by 40+ couples.

Autonomous Drone Platform for IESL-RoboGames

[IESL-RoboGames](#)

Docker, Pixhawk, ArduPilot, Raspberry Pi, SSH

N1K0 — Micromouse Robot

[github.com/N1K0-micromouse](#)

ESP32, Custom PCB Design, Real-Time Sensor Fusion

Visit [github.com/sheronjay](#) to view my other projects.

Leadership & Volunteering

Selection Committee Lead — SLIoT 2026

- Headed participant selection and evaluation for a National level IoT and innovation-focused program, ensuring fair and technical assessment standards.

Program Committee Lead — Hit the Grounds 2025

- Led the planning and execution of an industry-facing competition with participation from 20 leading companies.

Technical Committee Member — IESL-RoboGames 2026

- Contributed to developing a containerized sandbox environment for autonomous drone development using Docker, supporting Pixhawk, ArduPilot, and Raspberry Pi based workflows.

Member — AIESEC in Colombo South (2025–2026)

- Contributed to cross-cultural experience programs and internal operations within a global student-led organization.

Interests & Achievements

- Contributor: Asia-Pacific Robot Contest (2025); Winner: Hitachi High-Tech Award
- Sports: Volleyball (All-Island Champion 2019), Cricket (2012–2022)
- Finalist: Devthon 3.0 (2026)
- Finalist: CodeRally (2025)
- Finalist: CyberShield 4.0 (2025)

Skills

Programming: C, C++, JavaScript, Python, SQL, Bash

Web/Frameworks: REST API, React, Angular, Playwright, Spring Boot, Laravel, SvelteKit, Express

Cloud & DevOps: Git, Docker, Kubernetes, ArgoCD, Terraform, Ansible, CI/CD, AWS (S3, EC2)

Languages: English (Professional working proficiency), Sinhala (Native proficiency)